LOOMBENCH BRIEF · GROUP COMPOSITION · SEE ALSO THE BRIEF ON OBJECTIVES

Substituting one agent's disposition in a homogeneous group of five LLM agents

Question. When five agents given the same disposition divide a shared resource, how does the final allocation differ if one of them is given another disposition instead?

Design. Five agents each start with 20% of a resource and act for 26 rounds: maintain, claim a neighbour's part, concede, or propose joint ownership; the neighbour accepts, refuses or counters. An agent holding 46% is offered no further claims (the claim threshold). Each agent's prompt holds a one-sentence disposition and no objective. At k = 0 all five agents have disposition A; at k = 1 one of them has disposition B, and so on to k = 5. Runs are paired by seed, a match number that gives the same starting set-up; the substituted position rotates over seeds. 10 seeds for k = 0 and 1, 5 for k = 2 to 5; each run executed once. Decision systems: Jev (jev-1.13.0); GPT (Decisions) (gpt-6-luna, OpenAI's Decisions API); GPT (chat) (gpt-6-luna as a chat model stating a probability for each option).

Result. Mean final share of the focal agent. Blunt among calm: 42.6% to 44.2% at k = 1 against 16.8% to 19.6% at k = 0; rude among calm: 38.8% to 46.6% at k = 1 against 18.6% to 19.6% at k = 0; greedy among cooperative: 43.8% at k = 1 against 9.1% at k = 0. Jev and GPT (Decisions) did not clearly differ in the main contrast's change, and did differ in the focal agent's claims, the other agents' refusals and claims per agent at k = 0.

Focal agent: the agent whose disposition is changed. **Other agents**: the four whose disposition is not, reported as their mean.
Share: the percentage of the resource an agent holds alone at the end; a part under joint ownership counts for neither holder. **Seed**: a match number, used to pair runs.
k: the number of agents with the second disposition. **Intervals**: Student-t 95% over seeds.

calm "Jessica is calm and relaxed, hard to upset, and keeps an even mood under stress."
blunt "Jessica is blunt and hard-headed, suspicious of people's motives, and does not mind starting an argument."
rude "Jessica is rude and inconsiderate, talks over people and talks down to them, and does not care if she gives offence."
cooperative "Jessica is cooperative and fair-minded, likes to work with others toward what suits everyone, and is glad to share."
greedy "Jessica is greedy and self-interested, wants more than her share, and takes what she can get."

FIGURE 1 **Final allocation in one seed as k agents are given the second disposition**

Each picture shows who held which part of the resource in each round (time runs bottom to top), one colour for each agent; hatching marks a design adopted from another agent, two-colour bands a part under joint ownership. Squares give the seating, outlined for agents with the second disposition, whose final shares are beneath (k = 1 to 4). The seed shown is the one closest to the mean result at k = 1.



FIGURE 2 **Final share at k = 0 and k = 1, by seed**

Each line is one seed (10 in each panel); dots and numbers are means. Δ focal is the mean paired change in the focal agent's share, in percentage points, with its 95% interval. The dotted line is the 46% claim threshold.

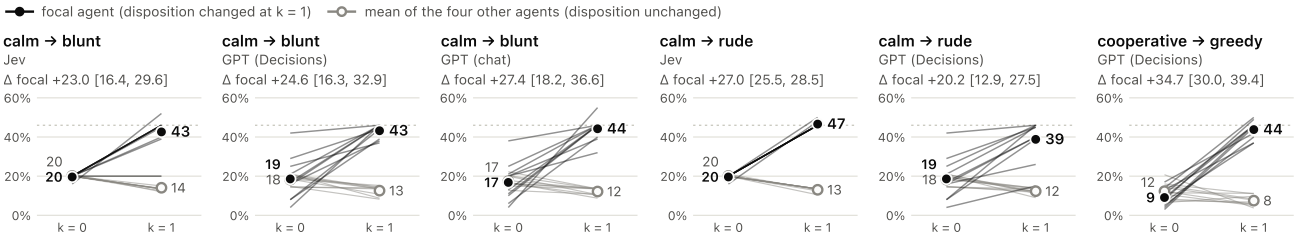


FIGURE 3 **The same contrast under each decision system**

Means with 95% intervals over seeds. Points and intervals share a scale within each column, and the vertical line marks zero. The last rows give paired differences between systems with their intervals; bold where the interval excludes zero.

calm → blunt	Focal agent: change in final share (pp)	Other agents: change in final share (pp)	Focal agent: claims per match, k = 1	Other agents: refusals per agent per match, k = 1	All agents: claims per agent per match, k = 0	Resource under joint ownership, k = 0 (%)
Jev	+23.0	−6.0	4.3	0.0	0.0	0.0
GPT (Decisions)	+24.6	−5.7	10.3	0.8	1.5	8.4
GPT (chat)	+27.4	−5.3	7.5	0.6	1.1	13.6
GPT (Decisions) − Jev paired, 10 seeds	+1.6 [-10.8, 14.0]	+0.4 [-2.9, 3.6]	+6.0 [3.9, 8.1]	+0.8 [0.5, 1.2]	+1.5 [1.1, 1.9]	+8.4 [3.6, 13.2]
GPT (chat) − Jev paired, 10 seeds	+4.4 [-4.5, 13.3]	+0.8 [-1.4, 2.9]	+3.2 [1.5, 4.9]	+0.6 [0.3, 0.8]	+1.1 [0.8, 1.5]	+13.6 [6.4, 20.8]

Findings and limitations

1. **Change in the focal agent's final share from k = 0 to k = 1 (Figure 2).** Mean paired change in percentage points over 10 seeds: calm → blunt, +23.0 to +27.4 across 3 systems, every interval excluding zero; calm → rude, +20.2 to +27.0 across 2 systems, every interval excluding zero; cooperative → greedy, +34.7 [30.0, 39.4] (GPT (Decisions)). Per other agent: −6.0 to −5.3 (calm → blunt); −7.0 to −5.9 (calm → rude); −4.8 [-8.7, -0.9] (cooperative → greedy).
2. **The claim threshold.** The rules stop offering an agent claims once it holds 46% of the resource. The focal agent finished at 45% or more in 5 to 10 of 10 seeds across the 6 comparisons in which its share rose. This is a threshold on the moves offered, not a limit on the final share, and these comparisons cannot show how much it affected the results.
3. **Jev and GPT (Decisions) compared (calm → blunt; Figure 3).** Paired over 10 seeds, GPT (Decisions) minus Jev: change in the focal agent's share +1.6 [-10.8, 14.0] pp, an interval that includes zero; claims by the focal agent 10.3 against 4.3 per match (+6.0 [3.9, 8.1]); refusals per other agent 0.8 against 0.0 (+0.8 [0.5, 1.2]). At k = 0, claims per agent per match were 1.5 against 0.0 (+1.5 [1.1, 1.9]). The change in share was not clearly different; the recorded actions were.
4. **Other comparisons.** Calm → rude, GPT (Decisions) minus Jev: −6.8 [-13.3, -0.3] pp in the focal agent's change (10 seeds). Cooperative → greedy, GPT (Decisions): 41.6% of the resource under joint ownership at k = 0 and 26.2% at k = 1 (−15.4 [-29.3, -1.5]).
5. **Group size (Jev, calm → blunt, 10 seeds each).** With 2, 3, 4 and 5 agents, each starting with 50, 33, 25, 20%, the focal agent ended with 71.6, 53.8, 44.6, 42.6% (claim thresholds 70, 53, 46, 46%): a change of +19.6 to +23.0 points. Each other agent's change was −21.6, −10.6, −6.5, −6.0 points. The threshold changes with group size, so the two cannot be separated here.
6. **Mean share of an agent with the second disposition, by k.** Jev: 43, 38, 29, 23, 20%; GPT (Decisions): 43, 41, 32, 25, 20% for k = 1 to 5 (5 seeds from k = 2). At k = 4 the one remaining agent of the group's kind held 8.4% (Jev) and 1.6% (GPT (Decisions)).

Limitations. Each run was executed once, so run-to-run variation is unmeasured; intervals describe variation over seeds. Each disposition is a single wording. A difference for the other agents is between the two resulting matches: it includes everything that followed from the substitution and does not isolate their direct response to the focal agent. A comparison between systems compares a model together with its interface and the way its probabilities are read, not models alone. The results concern this task, with its claim threshold, and these systems.

Every match and table · `node analysis/mix.mjs` prints the numbers.